

Euclid’s Elements — Book XI

Proposition 21

Formal Reconstruction — Technical Documentation

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Any solid angle is contained by plane angles which are together less than four right angles.

1 Purpose and Scope

Proposition XI.21 is the first proposition in the present reconstruction whose classical wording appears to demand a global arithmetic of several angles:

$$\angle BAC + \angle CAD + \angle DAB < 4R.$$

A literal transcription of Euclid’s proof would combine three applications of XI.20 with four applications of I.32 and then cancel six intermediate angles from a total of six right angles.

The formal development deliberately does not introduce a numerical angle measure, a general addition operation on angles, or a special “straight angle” object merely to reproduce that calculation. Instead it uses the same complement principle already familiar from the synthetic treatment of I.17: a bound by four right angles can be represented by comparing two supplementary angles with the third angle.

This leads to a considerably shorter proof than Euclid’s printed one. The final formal argument uses only:

- two ray extensions;
- Proposition XI.20;
- vertical angles;
- congruence transport;
- one line-in-plane closure argument preserving noncoplanarity.

Two independent production routes are retained. The direct Hilbert route uses `HilbertSpaceIncidence`; the Hilbert–Wyler route replaces the single spatial closure step by the corresponding field of `HilbertWylerAxioms`. Both routes share only the neutral target language from

2 Euclid's Statement and the Trihedral Scope

The classical statement is usually given in the unrestricted form:

Any solid angle is contained by plane angles whose sum is less than four right angles.

The proof, however, treats a solid angle bounded by three plane angles. Let A be the vertex and let the three edges be AB, AC, AD . The face angles are

$$\angle BAC, \quad \angle CAD, \quad \angle DAB.$$

The conclusion proved in the classical argument is

$$\angle BAC + \angle CAD + \angle DAB < 4R.$$

The Greek–English source used in the project explicitly notes that the printed proposition is proved only for the trihedral case; the extension to more than three face angles is stated to be straightforward. The current Lean theorem therefore formalizes exactly the trihedral case. It does not claim a general polyhedral-angle theorem.

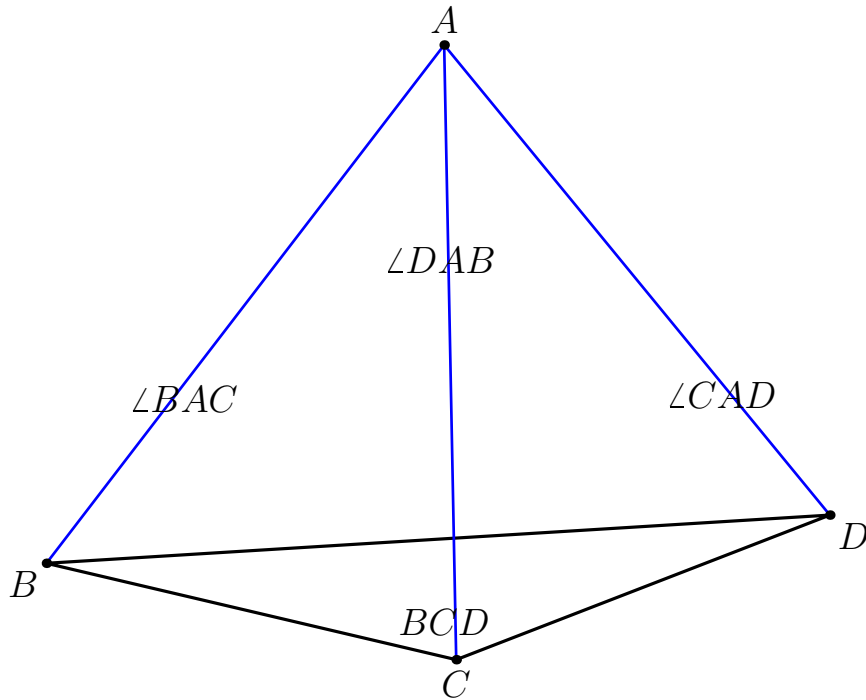


Figure 1: The trihedral configuration used in Proposition XI.21. The base points B, C, D are joined, while the three spatial edges from the vertex A determine the face angles $\angle BAC$, $\angle CAD$, and $\angle DAB$. The essential hypothesis is that A, B, C, D are not coplanar.

3 Euclid's Classical Proof

Euclid joins the three base points B, C, D . Proposition XI.20 is then applied at the three base vertices:

$$\begin{aligned}\angle CBA + \angle ABD &> \angle CBD, \\ \angle BCA + \angle ACD &> \angle BCD, \\ \angle CDA + \angle ADB &> \angle CDB.\end{aligned}$$

Hence the sum of the six angles

$$\angle CBA, \angle ABD, \angle BCA, \angle ACD, \angle CDA, \angle ADB$$

is greater than the sum of the three angles of triangle BCD .

By I.32,

$$\angle CBD + \angle BCD + \angle CDB = 2R.$$

Therefore those six angles are together greater than $2R$.

Euclid next applies I.32 to the three triangles

$$ABC, \quad ACD, \quad ADB.$$

Their nine angles are together equal to $6R$. Removing the six angles already shown to exceed $2R$, the remaining three angles at the vertex A are together less than $4R$:

$$\angle BAC + \angle CAD + \angle DAB < 4R.$$

The classical dependency graph is therefore

$$\begin{aligned}&\text{trihedral configuration } A; B, C, D \\&\quad \Downarrow \\&\text{XI.20 at } B, C, D \\&\quad \Downarrow \\&\text{six lateral/base angles exceed the three angles of } BCD \\&\quad \Downarrow \\&\text{I.32 on } BCD : \quad \text{base sum} = 2R \\&\quad \Downarrow \\&\text{I.32 on } ABC, ACD, ADB : \quad \text{nine-angle sum} = 6R \\&\quad \Downarrow \\&\angle BAC + \angle CAD + \angle DAB < 4R.\end{aligned}$$

4 Why the Arithmetic Proof Is Not Formalized Literally

The project treats an angle synthetically as a proper ray-pair configuration. A straight configuration is represented through betweenness and supplementary constructions rather than by enlarging the ordinary angle type to include a numerical angle of measure $2R$.

This is already visible in the formal treatment of I.17. A statement of the form

$$\alpha + \beta < 2R$$

is encoded by comparing α with a supplement of β , not by forming a numerical sum.

Likewise, I.32 in the present library records an exterior-angle decomposition. It is not a theorem about an additive monoid of angle measures. A literal formalization of Euclid’s XI.21 bookkeeping would therefore require substantial new infrastructure only to support a single cancellation argument.

During the XI.21 development such an infrastructure was deliberately postponed. The successful proof shows that it is not needed.

5 Neutral Synthetic Meaning of “Less Than Four Right Angles”

The common module

```
CGJteamLab/HilbertThreeAnglesFourRight.lean
```

introduces

```
def HilbertThreeAnglesLessThanFourRightAngles
  [HilbertIncidence Geo]
  [HilbertCongruence Geo]
  (A O B C P D E Q F : Geo.Point) : Prop :=
  Not (PrimCollinear Geo A O B) /\
  Not (PrimCollinear Geo C P D) /\
  Not (PrimCollinear Geo E Q F) /\
  exists X Y : Geo.Point,
    Geo.Between B O X /\
    Geo.Between D P Y /\
    HilbertTwoAnglesGreaterAngle
      Geo
      A O X
      C P Y
      E Q F
```

If the three original angles are denoted by

$$\alpha, \quad \beta, \quad \gamma,$$

then the two extensions construct their supplements

$$2R - \alpha, \quad 2R - \beta.$$

The formal certificate is

$$(2R - \alpha) + (2R - \beta) > \gamma,$$

which is the synthetic complement form of

$$\alpha + \beta + \gamma < 4R.$$

No angle-valued addition operation occurs in the definition. The only comparison object is the already established XI.20 relation

```
HilbertTwoAnglesGreaterAngle
```

For the trihedral proposition the public target is

```
def HilbertTrihedralAnglesLessThanFourRightAngles
  [HilbertIncidence Geo]
  [HilbertCongruence Geo]
  (A B C D : Geo.Point) : Prop :=
HilbertThreeAnglesLessThanFourRightAngles
  Geo
  C A D
  B A C
  D A B
```

The order CAD, BAC, DAB is intentional: it matches the comparison delivered directly by XI.20 after replacing one ray by its opposite.

6 The Direct Synthetic Shortcut

Extend the rays AC and AD through the vertex A :

$$C - A - X, \quad D - A - Y.$$

Thus

$$\angle BAX$$

is supplementary to $\angle BAC$, while

$$\angle CA Y$$

is supplementary to $\angle CAD$.

Now replace the ray AC by its opposite ray AX and reorder the trihedral configuration as

$$A; D, X, B.$$

Proposition XI.20 applied to this new trihedral configuration gives

$$\angle DAX + \angle XAB > \angle DAB.$$

Because $C - A - X$ and $D - A - Y$,

$$\angle DAX \cong \angle CA Y$$

by vertical angles. Since the project uses unoriented ordinary angles,

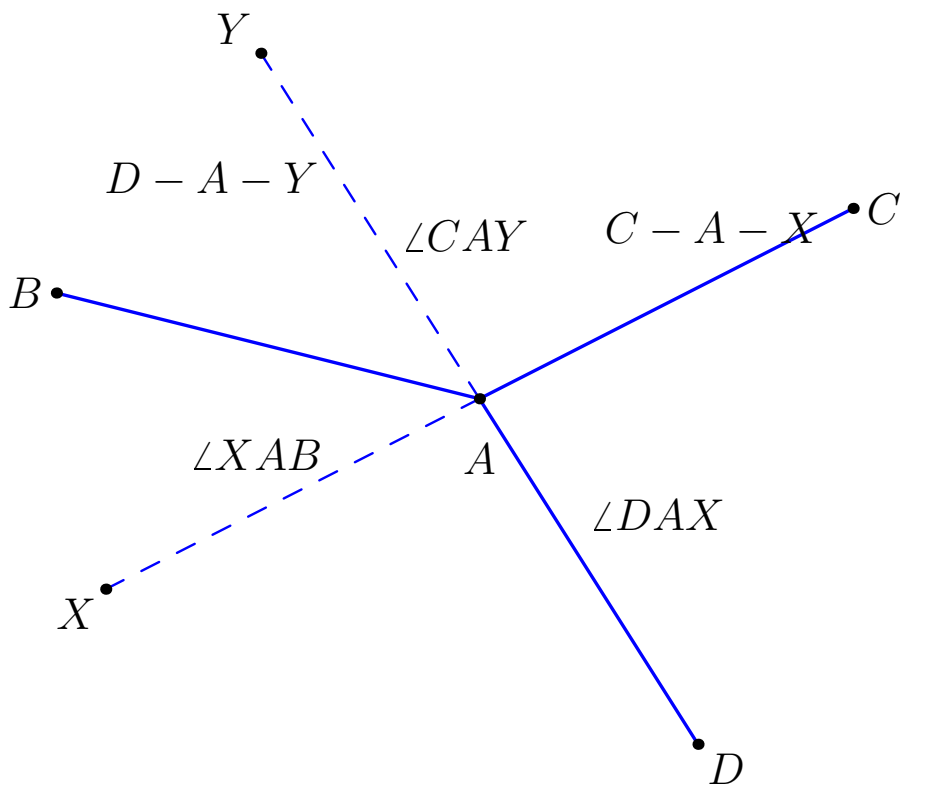
$$\angle XAB \cong \angle BAX.$$

Congruence transport therefore yields

$$\angle CA Y + \angle BAX > \angle DAB.$$

This is exactly the neutral complement certificate for

$$\angle CAD + \angle BAC + \angle DAB < 4R.$$



XI.20 on $A; D, X, B$: $\angle DAX + \angle XAB > \angle DAB$

Figure 2: The formal reduction of XI.21. The rays AC and AD are extended through A to X and Y . XI.20 is applied to the trihedral configuration $A; D, X, B$, giving $\angle DAX + \angle XAB > \angle DAB$. Vertical-angle and ray-order symmetries transport this to $\angle CAY + \angle BAX > \angle DAB$, the complement form of the desired four-right-angle bound.

The Lean proof graph is therefore much shorter than the classical angle-sum graph:

$$\begin{array}{c}
C - A - X, \quad D - A - Y \\
\Downarrow \\
A; D, X, B \text{ is trihedral} \\
\Downarrow \\
\text{XI.20} \\
\Downarrow \\
\angle DAX + \angle XAB > \angle DAB \\
\Downarrow \\
\angle DAX \cong \angle CAY, \quad \angle XAB \cong \angle BAX \\
\Downarrow \\
\angle CAY + \angle BAX > \angle DAB \\
\Downarrow \\
\angle BAC + \angle CAD + \angle DAB < 4R.
\end{array}$$

7 The Genuinely Spatial Obligation

The only nontrivial ambient-space obligation is to prove that replacing AC by the opposite ray AX preserves noncoplanarity.

Assume for contradiction that

$$A, B, X, D$$

lie in one plane π . Since $A \neq X$, the line AX lies in π . But C, A, X are collinear, so C lies on that same line. Consequently

$$C \in \pi.$$

Thus

$$A, B, C, D$$

would all lie in π , contradicting the original trihedral noncoplanarity.

$$C, A, X \text{ collinear} \implies C \in \pi$$

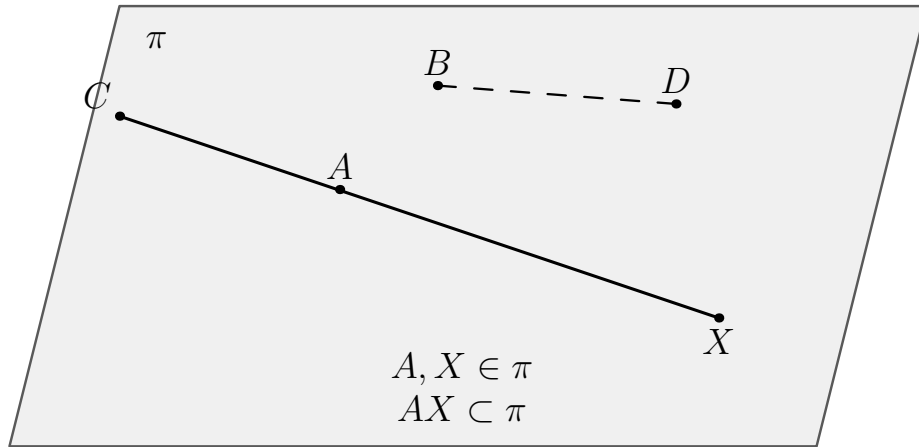


Figure 3: The sole spatial closure step in XI.21. If A, B, X, D lay in a plane π , then the line through the distinct points A, X would lie in π . Since C, A, X are collinear, this forces $C \in \pi$, contradicting the noncoplanarity of the original points A, B, C, D .

This is even weaker spatially than the corresponding XI.20 argument. XI.20 first had to construct a plane through three noncollinear points. XI.21 starts its contradiction with an already hypothesized plane and needs only line-in-plane closure.

8 Direct Hilbert 3D Reconstruction

After minimization of the inherited test binder, the direct production theorem is

```
theorem euclid_proposition_11_21
  [HilbertIncidence Geo]
  [S : HilbertSpacePrimitive Geo]
  [HilbertSpaceIncidence Geo]
  [HilbertCongruence Geo]
  (A B C D : Geo.Point)
  (hTri :
    HilbertTrihedralConfiguration
      Geo A B C D) :
  HilbertTrihedralAnglesLessThanFourRightAngles
    Geo A B C D
```

The key proposition-local lemmas are:

- hilbert_XI21_supplement_rays_exist;
- hilbert_XI21_trihedral_opposite_middle;
- hilbert_XI21_trihedral_reorder;
- hilbert_XI21_vertical_DAX_CAY;
- hilbert_XI21_angle_XAB_BAX;
- hilbert_XI21_direct_comparison.

The only Hilbert-space-specific call in the geometric core is

```
HilbertSpaceIncidence.line_in_plane
```

inside the proof that the opposite-ray trihedral configuration remains noncoplanar.

No use is made of I.32 in the production proof.

9 Hilbert–Wyler Reconstruction

The independent Wyler theorem is

```
theorem euclid_proposition_11_21_wyler
  [HilbertIncidence Geo]
  [S : HilbertSpacePrimitive Geo]
  [HilbertCongruence Geo]
```

```
[W : HilbertWylerAxioms Geo]
(A B C D : Geo.Point)
(hTri :
  HilbertTrihedralConfiguration
    Geo A B C D) :
HilbertTrihedralAnglesLessThanFourRightAngles
  Geo A B C D
```

Its production file is

```
CGJteamLab/Wyler/Proposition11_21.lean
```

The Wyler proof does not import or call `euclid_proposition_11_21`. It reconstructs the opposite-ray noncoplanarity lemma independently and calls the already independent Wyler version of XI.20:

```
euclid_proposition_11_20_wyler
```

In the sole spatial closure step the Wyler proof uses

```
W.line_in_plane
```

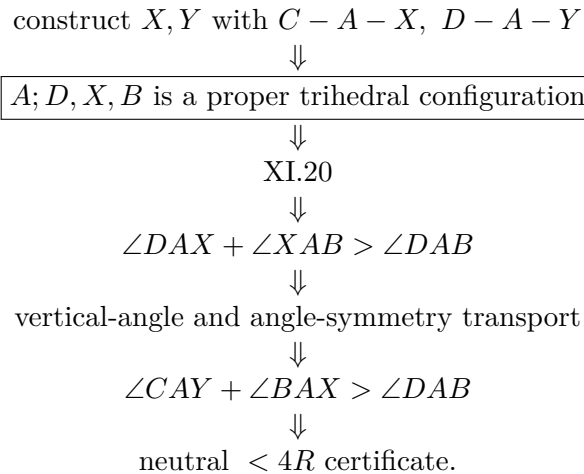
instead of

```
HilbertSpaceIncidence.line_in_plane
```

No `plane_through` field is needed in XI.21 itself. In particular, the proof does not use Wyler I.7, Smith I.5, span exchange, or an ambient dimension bound.

10 Hilbert versus Hilbert–Wyler Proof DAG

The two production routes are identical after the noncoplanarity-preservation step:



Only the boxed node differs. The direct route uses

```
HilbertSpaceIncidence.line_in_plane
```

whereas the Hilbert–Wyler route uses

```
HilbertWylerAxioms.line_in_plane
```

The common target and congruence-transport lemmas live in

```
HilbertThreeAnglesFourRight.lean
```

and are independent of the ambient spatial route.

11 E^n Transfer Audit

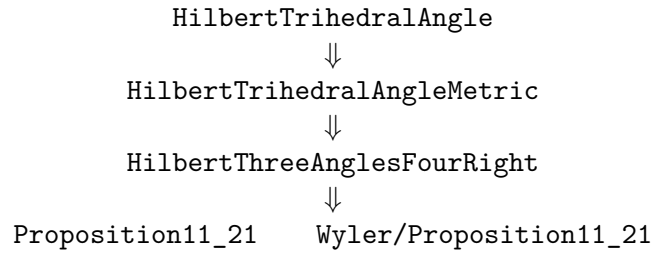
XI.21 gives an even cleaner dimensional-transfer result than XI.20.

1. **Purely planar/order step.** Ray extension, vertical angles, ordinary angle symmetry, congruence transport, and the complement representation are all dimension-free.
2. **Spatial step.** One must show that replacing one ray of a noncoplanar trihedral configuration by its opposite ray preserves noncoplanarity.
3. **Wyler replacement.** The proof needs only the principle that a plane containing two distinct points of a line contains the whole line.
4. **Expected E^4 and E^n analogue.** No local 3-flat is required. The same trihedral theorem applies to any ambient dimension in which the common point–line–plane incidence and planar metric theory are available.
5. **Missing higher-dimensional lemma.** For XI.21 there is none. The proposition is not an E^3 -specific theorem disguised by notation; the proof exposes it as a local dimension-free theorem about three rays.

This does not mean that all of Book XI is dimension-free. Rather, XI.20 and XI.21 now form a useful test pair showing where classical “solid geometry” can be promoted directly into the reusable synthetic library for later E^4 work.

12 Dependency Summary

The production architecture is



The two XI.21 branches additionally depend on their corresponding XI.20 production theorems:

euclid_proposition_11_20 euclid_proposition_11_20_wyler

The dependency classification is:

New proposition-specific geometric axiom	no
Numerical angle measure	no
General angle addition	no
I.32 in the production proof	no
Continuity	no new use
Hilbert Group IV	no, after binder minimization
Wyler I.7	no
Smith I.5 / exchange	no
Ambient dimension bound	no
Coordinates / vectors / scalar product	no
sorry/admit	no
Full repository build	successful, 2224 jobs

13 Conclusion

Proposition XI.21 initially appears to be the point where the project must introduce an arithmetic of finite angle sums. The formal reconstruction shows that this is unnecessary.

Euclid's printed proof remains mathematically important: it explains the classical relation between XI.20, the triangle angle theorem I.32, and the four-right-angle bound. But the synthetic complement representation reveals a shorter route. Two opposite-ray extensions transform the target into a single XI.20 comparison; vertical angles and unoriented-angle symmetry then transport that comparison directly to the required certificate.

The result has a second architectural consequence. The independent Wyler proof needs only line-in-plane closure at the ambient level. XI.21 therefore joins XI.20 as a reusable dimension-free theorem above the planar metric layer. In later E^4 work the natural task is not to reprove XI.21, but to show that the local Coxeter configuration satisfies the already available trihedral hypotheses.